

Chapter C3: Chemicals of the natural environment

Overview

Our way of life depends on a wide range of products made from natural resources. The Earth's crust provides us with metal ores and crude oil and our use of these impacts on the natural environment. Chemistry is fundamental to an understanding of the scale and significance of this human activity.

In Topic C3.1 the properties of metals are related to their structure and bonding, and in Topic C3.2 learners discover why the reactivity of a metal determines how it is extracted from its ores and

how new technologies enable us to extract metals from poor quality ores.

Electrolysis is explained in Topic C3.3, and learners learn about the wide variety of products made by electrolysis.

Finally, Topic C3.4 covers the separation of crude oil into fractions and the use of these fractions to make other chemicals and polymers. Within this context learners study the properties of simple molecules in relation to covalent bonding and intermolecular forces.

Learning about chemicals of the natural environment before GCSE (9–1)

From study at Key Stages 1 to 3 learners should:

- know the differences between atoms, elements and compounds
- be familiar with the use of chemical symbols and formulae for elements and compounds
- be familiar with the use of formulae and equations to represent chemical reactions
- understand chemical reactions as the rearrangement of atoms
- know about reactions of acids with metals to produce a salt plus hydrogen
- know some displacement reactions
- know the order of metals and carbon in the reactivity series
- know that carbon is used to obtain metals from metal oxides.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.

All other statements will be assessed in both Foundation and Higher Tier papers.

Learning about chemicals of the natural environment at GCSE (9–1)
